THE DEGREE OF A LOCALLY FREE MODULE ON A PROPER SCHEME OF DIMENSION AT MOST ONE IS THE MULTIPLICITY-WEIGHTED SUM OF ITS DEGREES ON THE ONE-DIMENSIONAL COMPONENTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites ten cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- Birational invariance of the degree for morphisms inducing isomorphisms at generic points of components. Stacks Project, Varieties, Lemma lemma-degree-birational-pullback (tag 0AYU). Reduces the general statement to the irreducible case via the coproduct morphism.
- Nilpotency of the ideal of a closed subscheme and length additivity at the generic point. Stacks Project, Cohomology of Schemes, Lemma coherent-lemma-power-ideal-kills-sheaf (tag 01YI); Algebra, Lemmas algebra-lemma-length-additive (tag 00IV) and algebra-lemma-dimension-is-length (tag 00IZ). Supply the nilpotency index and the length computation that drives the induction.

Source reference. Original reference 01W5: sources/stacks-chow/tags.txt, source lines 2470–2470. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01W3: sources/stacks-chow/tags.txt, source lines 2468–2468. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01RG: sources/stacks-chow/tags.txt, source lines 2301–2301. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYU: sources/stacks-chow/tags.txt, source lines 13802–13802. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01Y9: sources/stacks-chow/tags.txt, source lines 2548–2548. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 087T: sources/stacks-chow/tags.txt, source lines 10438–10438. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 00IV: sources/stacks-chow/tags.txt, source lines 693–693. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00IY: sources/stacks-chow/tags.txt, source lines 696–696. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00IX: sources/stacks-chow/tags.txt, source lines 695–695. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AYV: sources/stacks-chow/tags.txt, source lines 13803–13803. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let k be a field. Let X be a proper scheme of dimension ≤ 1 over k. Let $\mathcal{E}$ be a locally free $\mathcal{O}_X$-module of rank n. Then*

$$\deg(\mathcal{E}) = \sum m_i \deg(\mathcal{E}|_{C_i})$$

where $C_i \subset X$, $i = 1, \ldots, t$ are the irreducible components of dimension 1 with reduced induced scheme structure and m_i is the multiplicity of C_i in X.

Proof. Observe that the statement makes sense because $C_i \to \mathrm{Spec}(k)$ is proper of dimension 1; we will use below that any closed subscheme of X is proper over k by Morphisms, Lemmas 01W5 and 01W3. Consider the open subscheme $U_i = X \setminus (\bigcup_{j \neq i} C_j)$ and let $X_i \subset X$ be the scheme theoretic closure of U_i. Note that $X_i \cap U_i = U_i$ (scheme theoretically) and that $X_i \cap U_j = \emptyset$ (set theoretically) for $i \neq j$; this follows from the description of scheme theoretic closure in Morphisms, Lemma 01RG. Thus we may apply Lemma 0AYU to the morphism $X' = \coprod X_i \to X$. Since it is clear that $C_i \subset X_i$ (scheme theoretically) and that the multiplicity of C_i in X_i is equal to the multiplicity of C_i in X, we see that we reduce to the case discussed in the following paragraph.

Assume X is irreducible with generic point ξ. Let $C = X_{red}$ have multiplicity m. We have to show that $\deg(\mathcal{E}) = m \deg(\mathcal{E}|_C)$. Let $\mathcal{I} \subset \mathcal{O}_X$ be the ideal defining the closed subscheme C. Let $e \geq 0$ be minimal such that $\mathcal{I}^{e+1} = 0$ (Cohomology of Schemes, Lemma 01Y9). We argue by induction on e. If $e = 0$, then $X = C$ and the result is immediate. Otherwise we set $\mathcal{F} = \mathcal{I}^e$ viewed as a coherent $\mathcal{O}_C$-module (Cohomology of Schemes, Lemma 087T). Let $X' \subset X$ be the closed subscheme cut out by the coherent ideal $\mathcal{I}^e$ and let m' be the multiplicity of C in X'. Taking stalks at ξ of the short exact sequence

$$0 \to \mathcal{F} \to \mathcal{O}_X \to \mathcal{O}_{X'} \to 0$$

we find (use Algebra, Lemmas 00IV, 00IY, and 00IX) that

$$m = \mathrm{length}_{\mathcal{O}_{X,\xi}} \mathcal{O}_{X,\xi} = \dim_{\kappa(\xi)} \mathcal{F}_\xi + \mathrm{length}_{\mathcal{O}_{X',\xi}} \mathcal{O}_{X',\xi} = r + m'$$

where r is the rank of $\mathcal{F}$ as a coherent sheaf on C. Tensoring with $\mathcal{E}$ we obtain a short exact sequence

$$0 \to \mathcal{E}|_C \otimes \mathcal{F} \to \mathcal{E} \to \mathcal{E} \otimes \mathcal{O}_{X'} \to 0$$

By induction we have $\deg(\mathcal{E}|_{X'}) = m' \deg(\mathcal{E}|_C)$. By Lemma 0AYV we have $\chi(\mathcal{E}|_C \otimes \mathcal{F}) = r \deg(\mathcal{E}|_C) + n\chi(\mathcal{F})$. Putting everything together we obtain the result. □